L15

## Math Exploration 1

1. The acute angles of a right triangle are $a°$ and $b°$, where $a > b$ and both $a$ and $b$ are prime numbers. What is the least possible value of $b$? (2020 AMC 10B Problems, Question 4)

A. 2 B. 3 C. 5 D. 7 E. 11

## Math Exploration 2

4. A majority of the **30** students in Ms. Demeanor's class bought pencils at the school bookstore. Each of these students bought the same number of pencils, and this number was greater than **1**. The cost of a pencil in cents was greater than the number of pencils each student bought, and the total cost of all the pencils was **$17.71**. What was the cost of a pencil in cents? (2011 AMC 10A Problems, Question #10)

A. 7 B. 11 C. 17 D. 23 E. 77

7 Among three-digit numbers, there are _____ numbers with exactly 9 factors.

8 20! has $A$ factors, how many factors does $A$ have? _____

9 Using 2, 3, 4, 5, 6, 7 to form two three-digit numbers $A$ and $B$, then the maximum value of greatest common factor of the three numbers $A$, $B$, and 540 could be _____ .

10 A box contains no more than 200 pieces of candy. If the candy is taken out in groups of 2, 3, 4, or 6 pieces at a time, exactly one piece will be left in the box each time. If the candy is taken out in groups of 11 pieces, the box will be emptied completely. There are _____ pieces of candy in the box?

L16

## Math Exploration 2

3. What is the remainder of $19^{2024} - 8^{2024}$ divided by 9?

A. 0 B. 1 C. 2 D. 4 E. 5

## Math Exploration 2

5. Find the reaminder when $2025 \cdot 15^{2024}$ is divided by 4?

A. 0 B. 1 C. 2 D. 3 E. 4

2 There are a number of eggs in the box. If we take 3 each time, 1 is left in the end; if 5 are taken each time, 2 are left in the end; if we take 7 each time, 3 are left in the end, then there are at least _____ eggs in the box.

10 What is the remainder when $8 + 8 \times 8 + 8 \times 8 \times 8 + \cdots + \underbrace{8 \times 8 \times \cdots \times 8}_{2017}$ is dividing by 5?

A. 0 B. 1 C. 2 D. 3 E. 4

L17

## Math Exploration 1

2. A dessert chef prepares the dessert for every day of a week starting with Sunday. The dessert each day is either cake, pie, ice cream, or pudding. The same dessert may not be served two days in a row. There must be cake on Friday because of a birthday. How many different dessert menus for the week are possible? (2012 AMC 10B Problems, Question #11)

A. **729** B. **972** C. **1024** D. **2187** E. **2304**

## Math Exploration 2

**2.** The number **2013** has the property that its units digit is the sum of its other digits, that is $\mathbf{2+0+1=3}$. How many integers less than **2013** but greater than **1000** share this property? (2013 AMC10B Problems, Question #18)

A. **33** B. **34** C. **45** D. **46** E. **58**

# Math Exploration 2

**3.** How many of the base–ten numerals for the positive integers less than or equal to **2017** contain the digit **0**? (2017 AMC10B Problems, Question #16)

A. **469** B. **471** C. **475** D. **478** E. **481**

1 The side length of the three sides of a triangle is an integer, and the longest side length is 7 cm. How many types of triangles are there? _____

2 The sum of the digits on each digit of a three-digit odd number is 11. How many such three digits are there? _____.

( 2 ) 6 students line up, $A$ and $B$ must be next to each other, $B$ and $C$ cannot be next to each other. There are _____ ways to arrange.

## 4 Find the number of different permutations:

(4) 4 men and 4 women stand in a line. People of the same sex cannot be next to each other.

5 Seven students took pictures and found out how many standing methods there were under the following conditions:

(2) Seven people are lined up, and two classmates can't stand on the side. How many ways to stand?

(3) Tim and Bob have at least one person standing on the side. How many ways to stand?

6 Among the 10 students, 5 can install computers, 3 of them can install audio equipment, and the remaining 2 can install both computers and audio equipment. Today, the school decided to dispatch a 6-people installation team. In this team, 3 people are required to install computers, and 3 people are required to install audio equipment, so there are _____ different selection schemes.

8 12 identical apples should be divided into 3 days to eat, at least 2 on the first day, allow to eat no apple on the second day, and at least 3 on the third day, then there are _____ ways to eat.

L18

## Math Exploration 2

Ms. Carr asks her students to read any **5** of the **10** books on a reading list. Harold randomly selects **5** books from this list, and Betty does the same. What is the probability that there are exactly **2** books that they both select? (2020 AMC 10B Problems, Question #11)

A. $\frac{1}{8}$ B. $\frac{5}{36}$ C. $\frac{14}{45}$ D. $\frac{25}{63}$ E. $\frac{1}{2}$

## Math Exploration 2

An envelope contains eight bills: **2** ones, **2** fives, **2** tens, and **2** twenties. Two bills are drawn at random without replacement. What is the probability that their sum is **$20** or more? (2005 AMC 12B Problems, Question #11)

A. $\frac{1}{4}$ B. $\frac{2}{7}$ C. $\frac{3}{7}$ D. $\frac{1}{2}$ E. $\frac{2}{3}$

## Math Exploration 3

A box contains 5 chips, numbered 1, 2, 3, 4, and 5. Chips are drawn randomly one at a time without replacement until the sum of the values drawn exceeds 4. What is the probability that 3 draws are required? (2018 AMC 10B Problems, Question #6)

**A.** $\frac{1}{15}$ **B.** $\frac{1}{10}$ **C.** $\frac{1}{6}$ **D.** $\frac{1}{5}$ **E.** $\frac{1}{4}$

## Math Exploration 3

For a particular peculiar pair of dice, the probabilities of rolling 1, 2, 3, 4, 5 and 6 on each die are in the ratio 1 : 2 : 3 : 4 : 5 : 6. What is the probability of rolling a total of 7 on the two dice? (2006 AMC 12B Problems, Question #17)

**A.** $\frac{4}{63}$ **B.** $\frac{1}{8}$ **C.** $\frac{8}{63}$ **D.** $\frac{1}{6}$ **E.** $\frac{2}{7}$

## Math Exploration 3

A number is randomly selected from 1, 2, 3, ..., 30. What is the probability that it is an even number or divisible by 3?

**A.** $\frac{1}{6}$ **B.** $\frac{1}{3}$ **C.** $\frac{1}{2}$ **D.** $\frac{2}{3}$

## Math Exploration 4

Una rolls 6 standard 6-sided dice simultaneously and calculates the product of the 6 numbers obtained. What is the probability that the product is divisible by 4? (2021 Fall AMC 10B Problems, Question #14)

**A.** $\frac{3}{4}$ **B.** $\frac{57}{64}$ **C.** $\frac{59}{64}$ **D.** $\frac{187}{192}$ **E.** $\frac{63}{64}$

3 Five students $A$, $B$, $C$, $D$ and $E$ stand in a row. The probability that $A$ and $B$ are not adjacent to each other is _____ .

A $\frac{4}{5}$ B. $\frac{3}{5}$ C. $\frac{2}{5}$ D. $\frac{1}{5}$

6 If we pick two different numbers from $-2, -1, 0, 3, 4; 5$ and multiply them up, the probability that the product is a multiple of $5$ is _____ .

A. $\frac{2}{5}$ B. $\frac{3}{5}$ C. $\frac{1}{4}$ D. $\frac{1}{3}$ E. $\frac{1}{2}$

7 Holly writes down all the two-digit numbers which can be formed using the digits 1, 3, 7, and 9 (including 11, 33, 77 and 99). Warren selects one of these numbers at random. What is the probability that a prime number is selected?

A. $\frac{5}{8}$ B. $\frac{1}{2}$ C. $\frac{9}{16}$ D. $\frac{11}{16}$ E. $\frac{3}{4}$

9 Roll the dice twice. Let the numbers on the face be $m$ and $n$. What is the probability of the equation: $x^2 + mx + n = 0$ has integer solutions?

A. $\frac{5}{36}$ B. $\frac{7}{36}$ C. $\frac{1}{6}$ D. $\frac{5}{18}$ E. $\frac{1}{3}$

10 If a certain type of cell cannot divide, it dies. The probability of a cell either dying or dividing into two cells is $\frac{1}{2}$. Starting with two such cells, what is the probability that there are still living cells after two rounds of division?

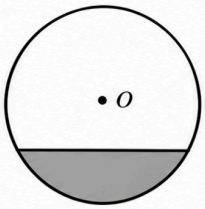
A. $\frac{39}{64}$ B. $\frac{25}{64}$ C. $\frac{31}{64}$ D. $\frac{29}{64}$

L19

## Math Exploration 1

2. A cylindrical container is put horizontally. The radius of the cross section is **1m**. If the depth of the water in the container is **0.4m**, then the width of the water is _____.

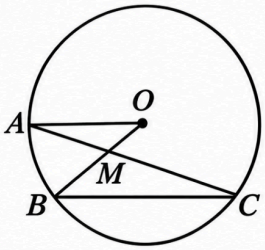
A. **0.8m** B. **1.2m** C. **1.6m** D. **1.8m**




## Math Exploration 2

2. As shown in the figure, in $\odot O$, chord $\overline{BC}$ is parallel to $\overline{OA}$. $\overline{AC}$ and $\overline{OB}$ intersect at $M$. $m\angle C = 20°$. Then $m\angle AMB$ is _____.

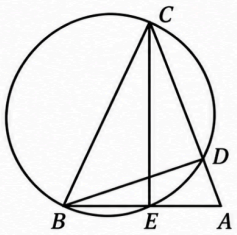
A. 30° B. 60° C. 50° D. 40°



## Math Exploration 3

4. Let $\triangle ABC$ be an isosceles triangle with $BC = AC$ and $m\angle ACB = 40°$. Construct the circle with diameter $\overline{BC}$, and let $D$ and $E$ be the other intersection points of the circle with the sides $\overline{AC}$ and $\overline{AB}$, respectively. Let $F$ be the intersection of the diagonals of the quadrilateral $BCDE$: What is the degree measure of $\angle BFC$? (2019 AMC 10A Problem, Question #13)

A. **90** B. **100** C. **105** D. **110** E. **120**




1 The number of incorrect statements is _____ .

①The major arc must be longer than the minor arc;

②Two circles with the same area are congruent circles;

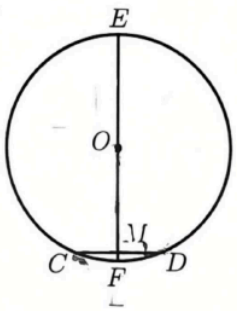
③The arcs with the same length are congruent arcs;

④Through a point inside the circle, we can draw infinite chords.

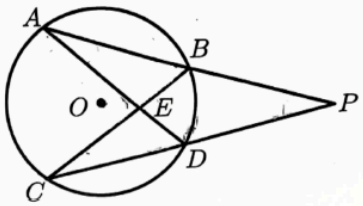
⑤Through a point inside the circle, we can draw infinite diameters.

A 1 B. 2 C. 3 D. 4

4 As shown in the figure, in $\odot O$, diameter $\overline{EF} \perp \overline{CD}$ at point $M$. If $CD = 2$, and $EM = 5$, then the radius of $\odot O$ is _____ .




9 As shown in the figure below, $\overline{AB}$ and $\overline{CD}$ are two chords of $\odot O$. $\overrightarrow{AB}$ and $\overrightarrow{CD}$ intersect at $P$. Connect $\overline{AD}$, $\overline{BC}$ and they intersect at $E$, $m\angle P = 30°$, $m\angle ABC = 50°$. Find $m\angle A$ and $m\angle AEC$.



L20

## Math Exploration 1

2. On the number line, number represented by point $A$ is 3 and number represented by point $B$ is $a$. The radius of $\odot A$ is 2. Which of the following is incorrect?

A. If $a < 5$, point $B$ is inside $\odot A$.

B. If $1 < a < 5$, point $B$ is inside $\odot A$.

C. If $a < 1$, point $B$ is outside $\odot A$.

D. If $a > 5$, point $B$ is outside $\odot A$.

## Math Exploration 2

1. On the coordinate plane, circle $O$ is centered at $(-3, 4)$ and the radius of it is 4, then _____.

A. $x$−axis intersects the circle at two points and $y$−axis is tangent to the circle.

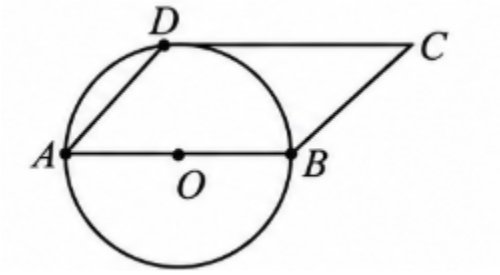
B. $x$−axis misses the circle and $y$−axis intersects the circle at two points.

C. $x$−axis is tangent to the circle and $y$−axis intersects the circle at two points.

D. $x$−axis is tangent to the circle and $y$−axis misses the circle.

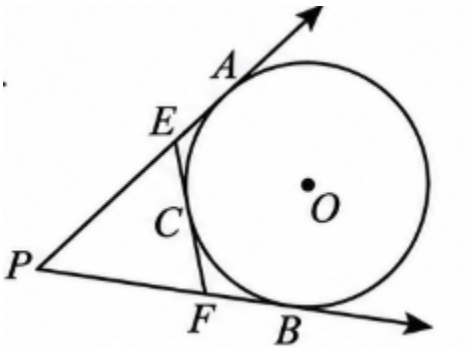
## Math Exploration 3

2. As shown in the figure below, quadrilateral $ABCD$ is a parallelogram, $m\angle C = 45°$. Draw $\odot O$ with $\overline{AB}$ being the diameter and it passes through $D$. Prove $\overline{CD}$ is tangent to $\odot O$.




## Math Exploration 5

2. As shown in the figure, $\overline{PA}$ and $\overline{PB}$ are tangent to the circle $O$ at points $A$ and $B$, respectively. $\overline{EF}$ is tangent to circle $O$ and intersects $\overline{PA}$ and $\overline{PB}$ at points $E$ and $F$, respectively. The tangent point $C$ lies on the arc $AB$. If the length of $\overline{PA}$ is 2, then the perimeter of triangle $PEF$ is .



Given that the radius of $\odot O$ is $4$, and $OP = 3$, then the positional relationship between $P$ and $\odot O$ is _____ .

A. $P$ is inside the circle
B. $P$ is on the circle
C. $P$ is outside the circle
D. Cannot be determined

4 In right triangle $\triangle ABC$, $m\angle C = 90°$, $AC = 4$, and $BC = 3$. Then the positional relationship between $\odot C$ whose radius is $2.5$ and $\overleftrightarrow{AB}$ is _____

5 Line $l$ intersects $\odot O$. The radius of $\odot O$ is $r$, and the distance from point $O$ to line $l$ is 6. Then the range of $r$ is _____ .

A. $r < 6$ B. $r = 6$ C. $r > 6$ D. $r \geqslant 6$

6 As shown in the figure, $\overline{AB}$ is the diameter of $\odot O$. $C$ is on $\odot O$, $D$ is on $\overrightarrow{AB}$, and $m\angle DCB = m\angle A$. Prove that $\overline{CD}$ is tangent to $\odot O$.

